Reg. No. | | | | | | | | | | | [14EC81]

EIGHTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

OPERATIONS RESEARCH AND MANAGEMENT

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions by choosing at least two questions from each part
2. Missing data may be suitably assumed.

PART - A

1. a. What are the applications of OR? **06**

b. Use the graphical method to solve the following LP problem. **08**

Maximize $Z = 6x_1+8x_2$

STC $5x_1+10x_2 \le 60$

$4x_1+4x_2 \le 40$

$x_1,x_2 \ge 0$

2. Use Big-M method to solve the following LP problem.

Maximize $Z = 2x_1+x_2+3x_3$ **14**

STC $x_1+x_2+2x_3 \le 5$

$2x_1+3x_2+4x_3 = 12$

and $x_1,x_2,x_3 \ge 0$

3. a. What are the rules for constructing the dual from primal? **07**

b. Obtain the dual of the following primal LP problem.

Maximize $Z = 3x_1+x_2+3x_3-x_4$

STC $2x_1-x_2+3x_3+x_4 = 1$ **07**

$x_1+x_2-x_3+x_4 = 3$

$x_1, x_2 \ge 0$ and x_3,x_4 unrestricted in sign.

4. a. Write a note on loops in transportation table and their properties. **05**

b. Obtain the initial basic solution to the transportation problem by VAM, for the details shown below: **09**

	D1	D2	D3	D4	Supply
S1	19	30	50	10	7
S2	70	30	40	60	9
S3	40	8	70	20	18
Demand	5	8	7	14	34

PART-B

5. a. List the various steps involved in Hungarian method. **06**

b. For M/s Lahari computer centre, assign the programmers to programs to have a minimum time in minutes. The following table gives the details: **08**

JAIN (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ramanagar-112

Programs		Programmers		
		X	Y	Z
	A	120	100	80
	B	80	90	110
	C	110	140	120

6. For the following details, find the optimum duration and the associated minimum project cost. (Take indirect cost as Rs 400/day). **14**

Activity	Immediate predecessor	Normal		Crash	
		Time (days)	Cost (Rs)	Time (days)	Cost (Rs)
A	-	10	20	7	30
B	-	8	15	6	20
C	B	5	8	4	14
D	B	6	11	4	15
E	B	8	9	5	15
F	E	5	5	4	8
G	A,D,C	12	3	8	4

7. a. Write the flow chart of game theory approach and explain. **08**

b. Solve the following game. What is the value of the game? **06**

Player A	Player B						
		B1	B2	B3	B4	B5	B6
	A1	3	-2	4	6	7	5
	A2	-1	8	2	4	12	6
	A3	16	8	6	14	12	7
	A4	1	11	-3	2	3	8

8. a. What are the essential features of queuing systems? Explain. **04**

b. The arrival rate of ships at a harbor follows Poisson distribution. The harbor has single duck to unload the containers from the incoming ships. The unloading time for the ships follows exponential (negative) distribution and hence the service rate also follows Poisson distribution. The arrival rate and the service rate are 8 ships per week and 14 ships per week respectively. **10**

Calculate:

i. Average number of waiting ships in the queue

ii. Average number of waiting ships in the system

iii. Average waiting time per ship in the queue

iv. Average waiting time per ship in the system

v. Utilization of the duck